DECK 15 · WEEK 4 · MORNING 3

# Duration, Convexity & Interest Rate Risk

*The heaviest math block of the course — duration measures, convexity, and curve-based risk for option-free and callable bonds*

CFA Level I · Vol 6 Learning Modules 10–13

# Roadmap — Fixed Income III: Duration, Convexity & Interest Rate Risk

*4 sections · ~2 hours · Built bottom-up from CFA readings*

**01**

### Interest rate risk & horizon yield

three sources of return · reinvestment vs price risk · BRWA VFO / Baywhite examples · duration gap

**02**

### Duration: Macaulay, modified, money, PVBP

MacDur weight tables · ModDur = MacDur/(1+r) · approximation · money duration · PVBP / DV01 · duration drivers

**03**

### Convexity & the convexity adjustment

second-order effect · $-MD \cdot \Delta y + \frac{1}{2} \cdot C \cdot \Delta y^2$ · money convexity · portfolio duration & convexity under parallel shifts

**04**

### Curve-based, effective & key rate

callables need effective duration · Viviyu negative convexity · key rate duration for curve shape · empirical vs analytical

# Learning objectives

*By the end of this deck you should be able to:*

1. Decompose the holding-period return on a fixed-rate bond into coupon, reinvestment, and capital-gain components, and explain how reinvestment and price risk offset each other at the Macaulay duration horizon.

2. Compute and interpret Macaulay, modified, money duration and PVBP for a coupon bond, and know when each is the right tool to reach for.

3. Predict how duration moves with coupon, yield, maturity, and time since the last coupon, and rank bonds on interest rate risk without a calculator.

4. Add a convexity adjustment to a duration-based price estimate and quantify, in both percent and currency terms, how much accuracy that second-order term buys.

5. Calculate portfolio duration and convexity as market-value-weighted averages, and state the key assumption (parallel yield-curve shift) and its limits.

6. Decide between yield duration, effective duration, key rate duration, and empirical duration for option-free bonds, callables / MBS, curve-shape views, and credit-risky portfolios.

SECTION 01

# Interest rate risk & horizon yield

# A fixed-rate bond's total return decomposes into three building blocks: coupon income, reinvestment income, and capital gain or loss on sale

*Horizon yield = the compounded annual rate that equates what you pay at t=0 to the total you end up with at horizon T (reinvested coupons + sale price).*

```
r_horizon  =  [ (FV_coupons + Sale_price)  /
                Purchase_price ]^(1/T)  -  1
```

*where: FV_coupons = future value of reinvested coupons at the reinvestment rate; Sale_price = bond price at horizon; T = years held*

**1** **Coupon & principal payments**
The cash itself — promised on scheduled dates. Drops out of interest rate risk as long as the issuer does not default.

**2** **Reinvestment of coupons**
Coupons earn the prevailing reinvestment rate until horizon. Rising rates help this leg; falling rates hurt it.

**3** **Capital gain / loss on sale**
Only if the bond is sold before maturity. Rising rates push the price down (loss); falling rates push it up (gain) — the opposite of reinvestment.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 10, "Sources of Return from Investing in a Fixed-Rate Bond", pp. 235–244.*

# Holding a bond to maturity strips out price risk but doubles down on reinvestment risk — the buy-and-hold investor wins from rate rises, loses from rate falls

*VFO buys BRWA 10-year 6.2% annual-coupon bond at par. Bond held to maturity → no capital gain / loss. Only reinvestment rate moves.*

| Scenario (rates shift immediately after purchase) | FV of coupons | Redemption | Horizon yield |
|---|---|---|---|
| Rates stable at 6.20% (base case) | 82.493 | 100.000 | 6.20% |
| Rates rise +100 bp → 7.20% | 86.475 | 100.000 | 6.43% |
| Rates fall −100 bp → 5.20% | 78.715 | 100.000 | 5.98% |

## WORKED — VFO HORIZON YIELD IF RATES RISE TO 7.20%

10-year 6.2% annual bond, purchase price 100, coupons reinvested at 7.2% for 10 years, redemption at par.

**1.** FV of coupons: =FV(7.2%, 10, 6.2, 0) = 86.475 per 100 par
**2.** Total ending cash: 86.475 + 100 = 186.475
**3.** Horizon yield: (186.475 / 100)^(1/10) − 1 = 6.43%

**Horizon yield = 6.43% (+23 bp vs base case — the gain is pure reinvestment benefit).**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 10, Examples 1–3 (VFO), pp. 239–242.*

# Selling before maturity exposes the investor to price risk that can swamp the reinvestment gain — Baywhite sells at year 4 and ends up losing on a rate rise

*Same BRWA 10-year 6.2% bond, but Baywhite has a 4-year horizon. On sale there are 6 years left to maturity.*

| Scenario (rate moves immediately after purchase) | FV of coupons (4y) | Sale price at t=4 | Horizon yield |
|---|---|---|---|
| Rates stable at 6.20% | 27.203 | 100.000 | 6.20% |
| Rates rise +100 bp → 7.20% | 27.609 | 95.263 | 5.28% |
| Rates fall −100 bp → 5.20% | 26.802 | 105.043 | 7.16% |

## WORKED — BAYWHITE WITH RATES UP 100 BP

Coupons reinvested at 7.2% for 4 years (FV = 27.609). Bond sold at t = 4 with 6 years to go, priced off 7.2% YTM.

**1.** Sale price: =PV(7.2%, 6, 6.2, 100) = 95.263 per 100 par (capital loss 4.737)
**2.** Total ending cash: 27.609 + 95.263 = 122.872
**3.** Horizon yield: (122.872 / 100)^(1/4) − 1 = 5.28%

**Horizon yield 5.28% — reinvestment gain of 0.406 is swamped by the 4.737 capital loss.**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 10, Examples 2 & 4 (Baywhite), pp. 240–242.*

# The duration gap — Macaulay duration minus horizon — tells you which side of interest rate risk dominates for a given investor

*At horizon = MacDur the two offset exactly. Shorter horizon: price risk dominates. Longer horizon: reinvestment risk dominates.*

**Duration gap = Macaulay duration − Investment horizon**

*where: Gap > 0 → price risk dominates (rising rates hurt), Gap < 0 → reinvestment risk dominates (falling rates hurt), Gap = 0 → nearly hedged.*

| Investor | Horizon | MacDur (bond) | Duration gap | Dominant risk |
|---|---|---|---|---|
| VFO (buy-and-hold) | 10y | 7.74y | −2.26 (negative) | Falling rates (reinvestment) |
| Hightest (8y horizon) | 8y | 7.74y | −0.26 (near zero) | Largely hedged |
| Baywhite (4y horizon) | 4y | 7.74y | +3.74 (positive) | Rising rates (price) |

**IMMUNISATION TAKEAWAY**

Matching the Macaulay duration of a bond (or portfolio) to the investor's horizon produces a single-period immunised position against a one-time parallel shift in the yield curve.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 10, §3 "Investment Horizon & Interest Rate Risk", pp. 244–250.*

SECTION 02

# Duration: Macaulay, modified, money & PVBP

# Macaulay duration is the present-value-weighted average time-to-receipt of a bond's cash flows, expressed in years

*Weights sum to 1 because the sum of the PV of cash flows equals the full bond price. MacDur is in units of periods; divide by periods per year to annualise.*

$$\text{MacDur} = \Sigma\ t \times [\ PV(CF_t) / PV_full\ ]$$

$$\text{Annualised MacDur} = \text{MacDur} / m$$

*where: t = time to each cash flow in periods; PV(CF_t) = present value of that cash flow; PV_full = dirty price; m = coupon periods per year.*

## ZERO-COUPON BONDS

Only one cash flow at maturity with weight 1.0. Macaulay duration = time-to-maturity exactly.

A 5-year zero has MacDur = 5y regardless of yield.

## PERPETUAL BONDS

No maturity, just infinite coupons. Closed form: MacDur = (1+r) / r.

At r = 6%, a perpetual has MacDur = 1.06 / 0.06 = 17.67y — longest duration of any plain-vanilla fixed income.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 10, §3 / Learning Module 11, §1, pp. 248–266.*

# Macaulay duration worked example — the 5-year 3.2% semi-annual BRWA bond has an annualised Macaulay duration of 4.66 years

*Bond priced at par so weight of each CF = its PV directly. Ten semi-annual periods of 1.6 coupon + 100 face at t=10. Periodic YTM = 1.6%.*

| Period t | Cash flow | PV @ 1.6% | Weight | t × Weight |
|---|---|---|---|---|
| 1–9 (coupons) | 1.6 each | 1.57 … 1.39 | 0.0157 … 0.0139 | 0.016 … 0.125 |
| 10 (coupon + par) | 101.6 | 86.69 | 0.8669 | 8.669 |
| Totals | — | 100.000 | 1.0000 | 9.320 |

## COMPUTE MACDUR (IN PERIODS, THEN ANNUALISED)

Sum of (time × weight) across all 10 periods = 9.3203 semi-annual periods. Periods per year m = 2.

**1.** Semi-annual MacDur = 9.3203 periods
**2.** Annualised MacDur = 9.3203 / 2 = 4.6601 years
**3.** Sanity check: Macaulay duration of a 5y par bond is always a bit less than 5y — a 6% coupon par bond has MacDur ≈ 4.44y, lower-coupon 3.2% lands a bit higher (as expected).

**Annualised Macaulay duration = 4.66 years.**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 11, Example 1, pp. 267–269.*

# Modified duration is Macaulay divided by (1 + r) and measures percentage price change per unit yield change — the first-order sensitivity

*Negative sign: price and yield move in opposite directions. Linear (tangent) approximation — accurate for small yield moves, off for large ones.*

$$\text{ModDur} = \text{MacDur} / (1 + r) \qquad \%\Delta PV_full \approx -\text{AnnModDur} \times \Delta y$$

*where: r = yield per period; Δy = change in annualised YTM (decimal). Linear estimate: doubles the error for larger |Δy| because price-yield curve is convex.*

**WORKED — BRWA 5y 3.2% BOND AT 3.2% YTM**

MacDur (periods) = 9.3203. Periodic YTM r = 0.016. Annualised modified duration sought.

**1.** ModDur (periods) = 9.3203 / (1 + 0.016) = 9.17351
**2.** AnnModDur = 9.17351 / 2 = 4.58676
**3.** Estimated %ΔPV_full for +80 bp YTM rise: −4.58676 × 0.0080 = −3.67%
**4.** Estimated %ΔPV_full for −80 bp YTM fall: −4.58676 × (−0.0080) = +3.67%

**AnnModDur = 4.587. A 100 bp YTM rise implies ≈ −4.59% price (before convexity).**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 11, §2 "Modified Duration", pp. 266–272.*

# Approximate modified duration bumps the yield up and down by a few basis points and reads the slope directly off the price curve

*The preferred method when MacDur is hard to obtain (contingent cash flows, default risk). Yields results essentially identical to the exact computation for small bumps.*

$$\text{AnnModDur} \approx (PV- - PV+) / [\, 2 \times \Delta\text{yield} \times PV_0 \,]$$

*where: PV_0 = current full price; PV+ / PV− = full prices after raising / lowering YTM by Δyield (annualised).*

**WORKED — BRWA 5y 3.2% BOND, 5 BP BUMP**

PV_0 = 100.00 (priced at par). Δyield = 0.0005 (5 bp). Compute PV+ at 3.25% YTM and PV− at 3.15% YTM.

**1.** PV+ at 3.25% = 99.770965 per 100 par
**2.** PV− at 3.15% = 100.229641 per 100 par
**3.** AnnModDur ≈ (100.229641 − 99.770965) / (2 × 0.0005 × 100.00)
**4.** = 0.458676 / 0.100000 = 4.58676

**Approximate AnnModDur = 4.587 — indistinguishable from the exact 4.58675 above.**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 11, §2 "Approximate Modified Duration", pp. 270–272.*

# Money duration converts modified duration into a currency figure — the dollar price change per unit yield move on an actual position

*US markets call this "dollar duration". Drops the percentage scaling so P&L attribution is in the local currency directly.*

$$\text{MoneyDur} = \text{AnnModDur} \times \text{PV_full} \qquad \Delta\text{PV_full} \approx -\text{MoneyDur} \times \Delta y$$

*where: PV_full can be in percent-of-par (per 100) or in the actual currency position. Output matches the input unit.*

**WORKED — USD 100M BRWA 5y BOND POSITION**

Par USD 100M. Settlement 57 days into first coupon period → PV_full = 100.504 per 100 par. AnnModDur = 4.43092 (settlement-adjusted).

**1.** Market value = USD 100,000,000 × 100.504/100 = USD 100,503,921
**2.** MoneyDur = 4.43092 × USD 100,503,921 = USD 445,324,719
**3.** Estimated P&L for +100 bp YTM move: −USD 445,324,719 × 0.0100 = −USD 4,453,247
**4.** Estimated P&L for −100 bp YTM move: +USD 4,453,247 (linear / pre-convexity)

**Money duration USD 445.3M. A 100 bp yield rise costs the investor ~USD 4.45M.**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 11, §3 "Money Duration", pp. 278–284.*

# PVBP (the "dollar value of an 01") is the currency price change per 1 basis point — the bond-market standard risk number for trading desks

*Trading desks typically quote risk in PVBP/DV01 because it scales linearly with notional and lines up with quoted bid-offer in basis points.*

$$PVBP = (PV- - PV+) / 2$$ where PV± are full prices at YTM ∓ 1 bp

*where: Per 100 of par or on the actual position notional. PVBP is essentially (MoneyDur × 0.0001) up to convexity rounding.*

**WORKED — BRWA 5y 3.2% BOND, PVBP PER 100 PAR**

PV_0 = 100.504. Bump YTM by ±1 bp (i.e. 3.19% and 3.21%). Use PRICE function for both.

**1.** PV+ at 3.21% YTM = 100.459400 per 100 par
**2.** PV− at 3.19% YTM = 100.548465 per 100 par
**3.** PVBP = (100.548465 − 100.459400) / 2 = 0.044532 per 100 par
**4.** On USD 30M par: DV01 ≈ USD 30,000,000 × 0.000445 = USD 13,359

**PVBP = 0.0445 per 100 par. A single basis point is worth ~USD 13k on a USD 30M BRWA position.**

[1] Also called "PV01" and, in the US, "DV01" (dollar value of 1 bp). BPV (basis-point value) is the same as money duration × 0.0001.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 11, §3 "Price Value of a Basis Point", pp. 278–284.*

# Duration's four drivers — coupon, yield, maturity, and position within the coupon period — each pull in a specific, predictable direction

*Lower coupon or yield shifts more weight onto distant cash flows, lengthening duration. Longer maturity does the same via the par payment.*

| Feature | Change | Effect on duration | Intuition |
|---|---|---|---|
| Coupon rate c | c ↑ | Duration ↓ | Higher early cash flows pull the PV-weighted average time shorter |
| Yield-to-maturity r | r ↑ | Duration ↓ | Higher discount rate deflates distant cash flows more → weights shift to near term |
| Time-to-maturity T | T ↑ | Duration ↑ (par/premium) | Longer life = more distant cash flows with meaningful weight (discount bonds can invert at very long T) |
| | | | Each day closer to the next |

**RANKING RULE OF THUMB**

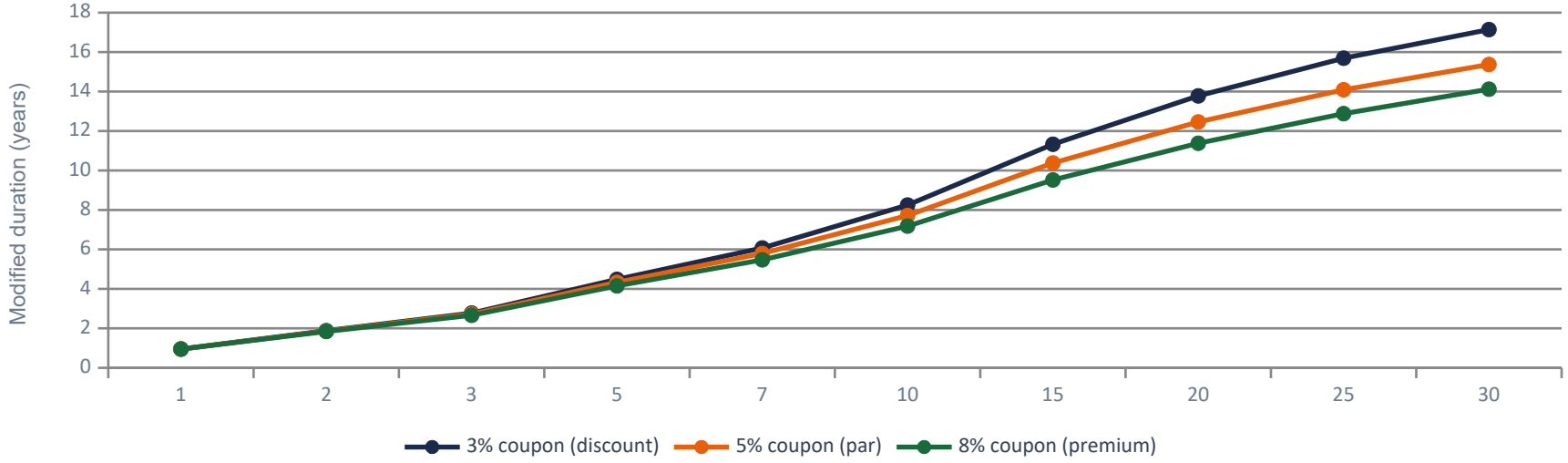
For two otherwise identical bonds: lower coupon → higher duration; lower yield → higher duration; longer maturity → higher duration (except deep-discount long bonds). Combine these and rank without a calculator.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 11, §4 "Properties of Duration", pp. 284–288.*

# Duration rises with maturity, falls with coupon, and falls with yield — visualised on one chart across 30 maturities

*All three curves share a 5% YTM. Par bond (c = r = 5%) sits between the two. At long maturities the curves asymptote toward (1+r)/r = 21y for 5% YTM.*

## Modified duration vs maturity at 5% YTM — three coupon levels




**READING THE CHART**

At any given maturity, lower-coupon bond sits above — confirming the coupon effect. All three curves converge toward (1+r)/r = 21y as maturity extends.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 11, Exhibit 5 "Properties of Macaulay Duration", p. 285.*

# Cross-section stress test: three bonds, same currency, different durations — the expected loss on a 100 bp move scales by more than 20×

*Chowdhury holds USD 30M par in each. Same 100 bp yield move — radically different dollar P&L because durations differ by order of magnitude.*

| Metric | BRWA 5y 3.2% | Australian 1y 0% | Romanian 30y 4.625% |
|---|---|---|---|
| Full price (per 100 par) | 100.815 | 99.262 | 107.429 |
| Annualised modified duration | 4.335 | 0.737 | 16.248 |
| PVBP (per 100 par) | 0.044 | 0.007 | 0.175 |
| Market value (USD 30M par) | 30,244,381 | 29,778,597 | 32,228,627 |
| Money duration (per 100 par) | 437.054 | 73.163 | 1745.518 |
| Expected loss, 100 bp YTM rise | −USD 1,311,163 | −USD 219,490 | −USD 5,236,553 |

**WHY THE ROMANIAN LOSS IS 24× LARGER**

MV difference is only ~8%; the driver is modified duration (16.25 vs 0.74). Investor expecting rates to rise would underweight the Romanian; expecting rates to fall, overweight it.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 11, §3 Question Set, pp. 281–284.*

SECTION 03

# Convexity & the convexity adjustment

# Duration is a linear approximation; real bond prices curve away from the tangent line — convexity measures that second-order effect

*For small Δy, duration alone is fine. For large Δy, the tangent line under-estimates the gain when rates fall and over-estimates the loss when rates rise — convexity corrects both.*

## DURATION (FIRST-ORDER)

Slope of the price-yield curve at the current YTM. Straight line. Works well for small Δy.

For Δy = 25 bp on a 5y bond, the error is negligible. For Δy = 100 bp on a 30y bond it's material.

## CONVEXITY (SECOND-ORDER)

Curvature of the price-yield curve. Always positive for an option-free bond.

Investors like convexity: for a given duration, more convexity = bigger upside from rate falls and smaller downside from rate rises. A free lunch in theory — so priced in.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 12, §2 "Bond Convexity", pp. 297–303.*

# The convexity-adjusted price change formula adds one-half times convexity times the squared yield change to the duration estimate

*Second-order Taylor expansion of price(yield). Always makes the estimated price higher (for option-free bonds) vs the linear estimate — regardless of direction.*

$$\%\Delta PV_full \approx -AnnModDur \times \Delta y + \tfrac{1}{2} \times AnnConvexity \times (\Delta y)^2$$

*where: Δy = change in annualised YTM (decimal). First term is duration (linear). Second is the convexity adjustment (always > 0 for option-free fixed-rate bonds).*

$$AnnConvexity \approx [\, PV- + PV+ - 2 \times PV_0 \,] \;/\; [\, (\Delta y)^2 \times PV_0 \,]$$

*where: Approximation via the same PV+ / PV− used for approximate modified duration — adds the sum instead of the difference.*

### LOWER COUPON

Pushes weight to the par payment → higher convexity.

### LONGER MATURITY

$(\Delta t)^2$ terms dominate → convexity rises faster than duration with T.

### LOWER YIELD

Discount factors closer to 1 → weights further out, higher curvature.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 12, Equation 1 & §2, pp. 297–300.*

# Worked example — BRWA 5y 3.2% bond: duration alone misses the 100 bp price change by 12 bp; the convexity adjustment closes the gap to less than 0.3 bp

*AnnModDur = 4.58676. AnnConvexity = 24.23895. Δy = ±0.01. Compare linear, convexity-adjusted, and exact PRICE-function values.*

| Δy | Exact %ΔPV | ModDur only | Err (bp) | ModDur + ½·Con·Δy² | Err (bp) |
|---|---|---|---|---|---|
| +100 bp | −4.46788% | −4.58676% | −11.89 | −4.46556% | +0.23 |
| −100 bp | +4.71035% | +4.58676% | −12.36 | +4.70795% | −0.24 |

**WALK-THROUGH — Δy = +100 BP**

Linear: −4.58676 × 0.01 = −4.587%. Convexity adjustment: ½ × 24.23895 × $(0.01)^2$ = +0.00121 = +12.1 bp.

**1.** Duration contribution: −4.58676 × 0.0100 = −0.045868 = −4.587%
**2.** Convexity contribution: ½ × 24.23895 × $(0.0100)^2$ = +0.001212 = +0.121%
**3.** Total estimate: −4.587% + 0.121% = −4.466%
**4.** Exact (PRICE function at 4.20% YTM): 95.53212 → −4.468% → error only 0.2 bp

**Duration + convexity lands within 0.3 bp of exact; duration alone off by 12 bp in both directions.**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 12, Exhibit 4, pp. 304–305.*

# Money convexity scales convexity to currency units — the extra dollar adjustment on a position's price change over what money duration predicts

*Same second-order logic, now in currency. Adds precision to P&L forecasting on large positions where the convexity term becomes material.*

$$\text{MoneyCon} = \text{AnnConvexity} \times \text{PV_full} \qquad \Delta\text{PV_full} \approx -\text{MoneyDur} \times \Delta y + \tfrac{1}{2} \times \text{MoneyCon} \times (\Delta y)^2$$

**WORKED — USD 100M BRWA POSITION, Δy = +100 BP**

AnnModDur = 4.58676, AnnConvexity = 24.23895, PV_full at par = USD 100,000,000.

**1.** MoneyDur = 4.58676 × 100,000,000 = USD 458,675,875
**2.** MoneyCon = 24.23895 × 100,000,000 = USD 2,423,894,503
**3.** Duration leg: −458,675,875 × 0.01 = −USD 4,586,759
**4.** Convexity leg: ½ × 2,423,894,503 × $(0.01)^2$ = +USD 121,195
**5.** Net estimate: −USD 4,465,564. Actual price drop: −USD 4,467,884. Error: −USD 2,320.

**Convexity term trims the duration-only over-estimate by ~USD 121k, closing 98% of the error.**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 12, Equations 3–4, pp. 304–305.*

# Convexity is always positive for an option-free bond and rises with maturity, falls with coupon and yield — the same drivers as duration but amplified

*Because the convexity formula involves t × (t+1), the maturity effect scales like $T^2$ — long bonds have enormously more convexity than short ones.*

| Bond | Maturity | Coupon | YTM | AnnModDur | AnnConvexity | Ratio Con / Dur |
|---|---|---|---|---|---|---|
| BRWA (semi) | 5y | 3.20% | 3.20% | 4.59 | 24.24 | 5.3 |
| Romanian (annual) | 30y | 4.625% | 4.75% | 15.91 | 369.64 | 23.2 |
| Gap | ×6 | — | — | ×3.5 | ×15.3 | ×4.4 |

## WHAT THIS MEANS FOR A LONG-BOND PORTFOLIO

A 100 bp yield shift on the Romanian bond is: duration piece −15.91%, convexity piece +1.85%, net −14.06%. The Romanian convexity term is 15× BRWA's — the convexity adjustment is no longer a tiny correction but a material component of the price change. For long-duration books, ignoring convexity is a systematic mis-estimation, not a rounding error.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 12, Example 1, pp. 300–301.*

# Portfolio duration is the market-value-weighted average of constituent durations — straightforward to compute, but assumes parallel yield curve shifts

*Weights are market values (not par, not face). Formula below extends identically to convexity. Bonds ought to share the same reporting currency.*

$$\text{AnnModDur_port} = \Sigma_i\ w_i \cdot \text{AnnModDur}_i \qquad \text{AnnConvexity_port} = \Sigma_i\ w_i \cdot \text{AnnConvexity}_i$$

*where: $w_i$ = MV(bond i) / MV(portfolio); weights sum to 1. Same price-change formula applies with these portfolio-level measures.*

**WORKED — BRWA + ROMANIAN TWO-BOND PORTFOLIO (USD 100M PAR EACH)**

BRWA MV = USD 50.0M (w = 0.5050), Romanian MV = USD 49.01M (w = 0.4950). Durations and convexities as on the previous slide.

**1.** Portfolio AnnModDur = 4.58676 × 0.5050 + 15.90637 × 0.4950 = 10.19004
**2.** Portfolio AnnConvexity = 24.23896 × 0.5050 + 369.64203 × 0.4950 = 195.21581
**3.** 100 bp parallel UP shift: %ΔPV = −10.190 × 0.01 + ½ × 195.216 × $(0.01)^2$ = −9.214%
**4.** A steepener (short rates up, long rates flat) would hit Romanian harder — MV-weighted duration understates this scenario.

**Portfolio value falls ~USD 9.12M on a 100 bp parallel rise in yields — provided the curve shifts parallel.**

[1] Theoretically correct method is to compute duration from the aggregate portfolio cash flows. MV-weighting is standard practice but baked-in to it is the parallel-shift assumption.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 12, §4 "Portfolio Duration and Convexity", pp. 307–311.*

# GBP 50M position, 10-year 3.50% bond — investor expects a 100 bp rate decline; the full duration-plus-convexity estimate is +8.78%, not +8.38% from duration alone

*For a 100 bp move the convexity term is 40 bp of the total estimate — large enough to matter for position sizing and stop-loss setting.*

$$\%\Delta PV_full = (-8.376 \times -0.01) + \tfrac{1}{2} \times 81.701 \times (-0.01)^2 = 0.08376 + 0.004085 = +8.78\%$$

**WORKED — GBP 50M POSITION · ANNMODDUR 8.376 · ANNCONVEXITY 81.701**

10y 3.50% semi-annual coupon bond, priced at par. Investor's view: −100 bp parallel rate decline. Compute both the percent price change and the money convexity adjustment.

**1.** Duration leg: −8.376 × (−0.01) = +0.08376 = +8.376%
**2.** Convexity leg: ½ × 81.701 × $(-0.01)^2$ = +0.004085 = +0.409%
**3.** Combined estimate: +8.78%
**4.** MoneyDur = 8.376 × GBP 50,000,000 = GBP 418,800,000
**5.** MoneyCon = 81.701 × GBP 50,000,000 = GBP 4,085,033,604
**6.** Money convexity adjustment: ½ × 4,085,033,604 × $(0.01)^2$ = +GBP 204,252

**Total expected gain ≈ GBP 4.39M (duration) + GBP 204k (convexity) = GBP 4.59M on a −100 bp move.**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 12, Self-assessment Q3–4, pp. 295–297.*

SECTION 04

# Curve-based, effective & key rate duration

# Yield-based duration assumes cash flows are certain — it breaks down for callable, puttable, or mortgage-backed securities whose cash flows depend on rates

*A callable bond has no single well-defined YTM because the issuer can redeem early at the call price if rates fall — so MacDur and ModDur lose meaning. Cue effective duration.*

## CALLABLE BOND

Issuer holds a call option. When rates FALL far enough, issuer redeems at call price → investor loses upside.

Curve: tangent flattens as yields fall, then curve becomes NEGATIVELY convex. Effective duration < straight-bond duration when rates are low.

## PUTABLE BOND

Investor holds the put option. When rates RISE far enough, investor sells back at par → downside truncated.

Curve: flattens on the rising-rate side. Effective duration < straight-bond duration when rates are high. Convexity stays positive.

## MBS / ABS

Prepayment option held by homeowners. Falling rates → refinance wave → early principal return → same flattening + negative convexity as callables.

Effective duration from an OAS model is the only usable risk number.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 13, §2 "Curve-Based Interest Rate Risk Measures", pp. 320–326.*

# Effective duration and effective convexity measure price sensitivity to a parallel shift of the benchmark yield curve — valid whether cash flows are certain or not

*Formulas look like the approximate-duration ones — but the bump is on the benchmark curve, not on the bond's own YTM. PV± come from an option-valuation model.*

$$\text{EffDur} = (PV- - PV+) / [\, 2 \times \Delta\text{Curve} \times PV_0 \,]$$

*where: ΔCurve = parallel shift of the benchmark (e.g. government par) curve. PV± re-priced with the full option-pricing model under the shifted curve.*

$$\text{EffCon} = [\, PV- + PV+ - 2 \times PV_0 \,] / [\, (\Delta\text{Curve})^2 \times PV_0 \,]$$

*where: Can be negative for callables / MBS. $\%\Delta PV_full = -\text{EffDur} \times \Delta\text{Curve} + \frac{1}{2} \times \text{EffCon} \times (\Delta\text{Curve})^2$.*

**WHY MODDUR ≠ EFFDUR (EVEN FOR OPTION-FREE BONDS)**

A parallel shift in the government PAR curve does not produce a parallel shift in the SPOT curve. The bond's own YTM moves by a slightly different amount than the par-curve bump. So EffDur and ModDur differ a little even for a plain bond. The gap shrinks when the yield curve is flatter, the maturity shorter, and the bond closer to par. Only a flat yield curve makes them identical.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 13, §2 Equations 1–2, pp. 322–324.*

# Callable bonds can have negative effective convexity when rates fall below the coupon — the issuer's call option truncates the upside

*PM targets duration between 7 and 8 with positive convexity. Callable bond passes the duration screen but fails on convexity — recommend not to buy.*

| Scenario | Curve move | Full price | Computation |
|---|---|---|---|
| Base case (today) | 0 | 101.060 | PV_0 — observed market price |
| Benchmark par curve +25 bp | +0.0025 | 99.050 | PV+ — from option pricing model |
| Benchmark par curve −25 bp | −0.0025 | 102.891 | PV− — from option pricing model |

## EFFECTIVE DURATION AND CONVEXITY

Plug PV_0 = 101.060, PV+ = 99.050, PV− = 102.891, ΔCurve = 0.0025 into the formulas.

**1.** EffDur = (102.891 − 99.050) / (2 × 0.0025 × 101.060) = 3.841 / 0.5053 = 7.601
**2.** EffCon = (102.891 + 99.050 − 2 × 101.060) / [$(0.0025)^2$ × 101.060]
**3.** = (−0.179) / 0.000631625 = −283.4
**4.** Interpretation: duration is in the PM's 7–8 range, but negative convexity means a falling-rate rally delivers LESS than duration suggests. Decline.

**EffDur = 7.60, EffCon = −283. Negative convexity — skip.**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 13, Example 1 "Viviyu Inc. callable bond", pp. 324–325.*

# Key rate duration (partial duration) is the bond's sensitivity to a change in one point on the benchmark curve — how the bond responds to a steepening or twist

*EffDur compresses the whole curve into one number. KRD breaks it apart by tenor, letting a manager diagnose "shaping risk" and tilt against a view of the curve shape.*

$$\text{KeyRateDur}_k = -(1 / PV) \times (\partial PV / \partial r_k) \qquad \Sigma_k \text{ KeyRateDur}_k = \text{EffDur}$$

*where: r_k = benchmark yield at the kth key maturity (e.g. 2y, 5y, 10y, 30y). Sum of key rate durations across the curve = effective duration (parallel shift).*

| Tenor | 2y bond | 5y bond | 10y bond | Portfolio (EW) |
|---|---|---|---|---|
| Key rate 2y | 1.990 | – | – | 0.711 |
| Key rate 5y | – | 4.938 | – | 1.675 |
| Key rate 10y | – | – | 9.828 | 2.981 |
| Sum = Effective duration | 1.990 | 4.938 | 9.828 | 5.368 |

**USE CASE**

With a steepening view, over-weight short-KRD bonds and under-weight long-KRD bonds; the portfolio loses less (or gains more) than a pure duration-matched index.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 13, §4 "Key Rate Duration", pp. 329–333.*

# Key rate duration worked example — a portfolio of two bonds facing a curve steepening: sell the bond whose key tenor suffers the most

*Curve steepens: short rates up 100 bp, 5y up 150 bp, 10y up 200 bp, 20y up 250 bp, 30y up 300 bp. Equal-weight portfolio of two bonds — keep one, sell the other.*

| Bond | Key tenor | KeyRateDur | Shift at tenor | Estimated %ΔPV |
|---|---|---|---|---|
| Bond A | 5y | 2.702 | +150 bp | −2.702 × 0.015 = −4.05% |
| Bond B | 10y | 3.953 | +200 bp | −3.953 × 0.020 = −7.91% |

**DECISION — WHICH BOND TO SELL**

Both bonds lose under the steepener; the question is magnitude. Use KRD × shift at each bond's key tenor.

**1.** Bond A: −2.702 × 0.0150 = −4.05%
**2.** Bond B: −3.953 × 0.0200 = −7.91%
**3.** Bond B's losses are ≈ 2× Bond A's — sell Bond B first.
**4.** Alternative action: pair-trade the steepener — receive 5y (long Bond A), pay 10y (short Bond B).

**Sell Bond B. KRD isolates the bond's exposure to the specific tenor that's moving most.**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 13, §4 Question Set, pp. 332–333.*

# Empirical duration replaces mathematical formulas with historical data; it's the right measure for credit-risky bonds in market stress, where yields and spreads move in opposite directions

*Analytical (MacDur, ModDur, EffDur, KRD) assumes yields and spreads are independent. Empirical regression picks up the real-world correlation — usually negative in flight-to-quality episodes.*

## ANALYTICAL DURATION

Formula-based from a bond's own cash flows or an OAS model. Assumes spreads uncorrelated with the benchmark.

Works well for: governments, high-IG corporates, benign regimes.

Fails when: credit spreads widen as benchmarks rally (e.g. Mar 2020 — the COVID flight-to-quality episode).

## EMPIRICAL DURATION

Regression of HISTORICAL bond returns on benchmark yield changes. Captures the spread-yield correlation directly.

For HY bonds during market stress, empirical duration is MATERIALLY LOWER than analytical — because widening spreads partly offset falling benchmarks on the bond price.

Challenges: factor choice, sample window, regime stability.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 13, §5 "Empirical Duration", pp. 333–336.*

KEY TAKEAWAYS

# Deck 15 — Fixed Income III: Duration, Convexity & Interest Rate Risk

- Bond total return = coupon + reinvested-coupon income + capital gain/loss on sale. Reinvestment and price risks are inverse in rate direction.
- Duration family: MacDur (time, in years) → ModDur = MacDur / (1+r) (% per Δy) → MoneyDur = ModDur × PV_full (currency per Δy) → PVBP = MoneyDur × 0.0001 (currency per 1 bp).
- Convexity adjustment $\%\Delta PV \approx -MD \cdot \Delta y + \frac{1}{2} \cdot C \cdot \Delta y^2$ closes the residual linearisation error; for BRWA at 100 bp it cuts the 12 bp duration-only error down to <0.3 bp.
- Effective duration / convexity are curve-based and valid for callables, puttables, MBS. Callables can go NEGATIVELY CONVEX when rates fall (Viviyu EffDur = 7.60, EffCon = −283).
- Duration gap = MacDur − horizon. Negative → falling-rate risk. Positive → rising-rate risk. Zero → nearly immunised against one-time parallel shifts.
- Duration drivers: lower coupon, lower yield, longer maturity → higher duration. Durations are reliable rankings without a calculator once those rules are internalised.
- Portfolio duration/convexity = MV-weighted averages. Fast and practical but implicitly assumes a PARALLEL curve shift — mask steepening / twisting risk.
- Key rate duration isolates exposure to individual tenors (sum = EffDur) and drives steepener/flattener trades. Empirical duration is the correct measure for credit-risky bonds in stress — yields and spreads become negatively correlated.